GENETICS AND GENOMICS

Correlated responses to long-term divergent selection for 8-week body weight in female White Plymouth Rock chickens: Sexual maturity

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ABSTRACT Reported here are correlated responses for reproductive traits to long-term divergent selection (54 generations) for 8-week body weight (BW8). Comparisons involved both selected and relaxed lines. Traits measured were age at first egg (AFE), body weight at first egg (WFE), and ratio of body weight and age at first egg (WAFE). Although sexual maturity was delayed in the selected lines, the effect was more pronounced in the low than high selected and relaxed lines. Selection for low BW resulted in decreases in WFE and WAFE. Correlated responses to selection for high BW were increased WFE and WAFE. Minimum AFE, WFE, and WAFE in relation to sexual maturity were line specific and influenced by selection for BW8. WAFE provided a "yardstick" for target body weights that were optimum for successful attainment of sexual maturity and higher reproductive rates. Such may be line specific. There was opposition between relaxed and artificial selection, resulting in a higher reproductive performance and fitness for the former.

Key words: chickens, weight at sexual maturity, age at sexual maturity, selection, correlated response

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INTRODUCTION

Long-term divergent selection for 8-week body weight (**BW8**) is accompanied by changes in reproductive performance. Correlated responses associated with selection for high and low BW from an earlier generation period (Generation 0 to 27) included delayed sexual maturity and lower egg production (Siegel and Dunnington, 1985). These were a consequence of negative correlations between growth and reproductive traits. A strong negative correlation exists between body weight and reproductive efficiency because of selection for increased growth (McCarthy and Siegel, 1983; Decuypere et al., 2010). The delayed sexual maturity in low-weight chickens may have been associated with a decrease in energy reserves integral to egg production, a view consistent with the delay in age at sexual maturity observed for this line (Siegel and Dunnington, 1987).

Sexual maturity and its onset in an animal's reproductive life are of profound importance from both evolutionary and economic perspectives (Wright et al., 2012). Chronological age, BW, and body composition influence onset of sexual maturity (Dunnington and Siegel, 1985; Eitan and Soller, 2001, 2009). Namely, for a chicken to commence egg production, there must be specific body composition, minimum age, and minimum BW, which are line specific (Dunnington et al., 1983, 1984; Brody et al., 1984; Zelenka et al., 1986a,b, 1987). The QTL analysis by Podisi et al. (2011) revealed overlapping QTL for body weight at first egg and at 9 wk of age indicated that QTL affecting growth also affected body weight at first egg. Co-localization of QTL for body weight, growth, and sexual maturity suggests that body weight and growth are closely related to the attainment of sexual maturity and that the genetic factors associated with growth have correlated effects on puberty.

There are dramatic changes in relationships between a selected trait and traits associated with fitness, which imply an opposition between artificial and natural selection that demonstrates a strong effect of the latter on the traits, regardless of how long artificial selection has been practiced.

The objectives of this study are (i) to measure the correlated responses of long-term (54 generations) selection for BW8 on sexual maturity, (ii) to determine the response of body weight at first egg from generation 27 to 54 and relationships of growth to sexual maturity, and (iii) to compare sexual maturity of selected and relaxed lines.

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MATERIALS AND METHODS

Experimental Design

Procedures for This Study Were Approved by the Institutional Animal Care and Use Committee at Virginia Tech. Fifty-four generations of selection were conducted for high or low BW8 in chickens. The foundation stock consisted of crosses of 7 moderately inbred lines of White Plymouth Rock (**WPR**) chickens (Siegel, 1962). From this segregating gene pool, chickens with heavier BW8 were chosen as parents for the high line (**HWS**) and those with lighter BW8 were selected as parents for the low line (**LWS**). Individual selection was then practiced within the respective HWS and LWS lines in all subsequent generations, with BW8 as the single criterion of selection. Numbers of sires and dams selected to produce each line were 8 and 48 through the fourth generation of selection (S4), 12 and 48 from S5 to S25, and 14 and 56 after S25. These parents are selected from within-line groups of approximately 150 to 250 individuals, depending on generation. Pure truncation selection was not used, however, so that overrepresentation of sire and dam families would be avoided (Dunnington and Siegel, 1996; Dunnington et al., 2013). This effort to curtail inbreeding from common ancestry and avoid matings among half sib or close relatives and to avoid selection of particular sire or dam families reduced, yet could not avoid, inbreeding from common ancestry and resulted in similar population structures in the lines. By generation 48, inbreeding in both lines was moderately high, and effective population size was low, as expected, in closed lines (Marquez et al., 2010).

Relaxed Lines In S6, S13, S19, S26, S34, and S43, chickens from lines HWS and LWS were chosen at random before selection of breeders for the selected lines to establish sublines in which selection for BW8 was relaxed. Each subline (designated **HWR**1, HWR2, HWR3, HWR4, HWR5, and HWR6 for line HWS and **LWR**1, LWR2, LWR3, LWR4, LWR5, and LWR6 for line LWS) was maintained as a contemporary of its corresponding selected line and reproduced by random matings each generation. The relaxed lines were produced by using pooled semen within a subline to artificially inseminate females in that line. The same husbandry procedures and environments were employed for the relaxed lines as those practiced in the selected lines. A relaxed line was maintained for 7 to 11 generations. Each subsequently relaxed line was initiated one generation before the previous relaxed line was discontinued (Dunnington et al., 2013; Jambui et al., 2017).

Husbandry Each generation, chicks were hatched on the first Tuesday in March of every year and wing-banded for individual identification. If adequate numbers were not obtained from this hatch, a second hatch was produced on the third Tuesday in March. Every generation, chicks were reared in identical pens with concrete floors, wood shavings as bedding, continuous light, and hot air brooding until 8 wk of age (Dunnington and Siegel, 1996; Dunnington et al., 2013). From 8 to 18 wk, they were exposed to natural photoperiod, after which they were provided artificial light from 0600 to 2000 h in a light-controlled environment (Liu et al., 1995). Other than routine husbandry in a rather isolated environment, the only disease preventatives were the addition of a coccidiostat to the starter and developer diets and vaccination for Marek's disease, which commenced after generation 17 of selection. Through generation 13, chickens were maintained in littered floor pens. Thereafter, they were reared on litter to approximately 126 d of age and then individually housed in wire cages. Throughout the study, feed in mash form and water were provided ad libitum with the same formulations each generation. Chicks were fed the starter until 8 wk of age, developer from 8 to 20 wk, and a breeder diet thereafter. Percentages of protein were 20, 16, and 16, and levels of metabolizable energy (**ME**) were 2,685, 2,761, and 2,772 kcal/kg in the starter, developer, and breeder diets, respectively. Beginning in generation 18, feed intake after 56 d was restricted for HWS and its relaxed sublines because of increasing difficulties with reproduction due to obesity (Dunnington and Siegel, 1985, 1996; Dunnington et al., 2013). The degree of restriction was adjusted periodically with age from 60 g at 57 d to 110 g by 147 days.

Traits Traits reported here are age at first egg (**AFE**), body weight at first egg (**WFE**), and the ratio of body weight and age at first egg (**WAFE**). Data for the WFE and WAFE were collected only since the generation 27 of selection.

The AFE was defined as age when the first egg was laid, provided that a second egg was laid within the next 9 days. In the later generations of selection, AFE in LWS was biased downward because an increasing number of pullets did not begin to produce eggs by 240 d of age (Dunnington and Siegel, 1985). Subsequently, the period was extended to 275 d of age starting in generation 37.

WFE (to the nearest 10 g) was obtained on the first d of lay starting in generation 27. WAFE was calculated by dividing the mean of WFE by AFE for each generation. This ratio reflects growth to sexual maturity and assumes that the dynamics of body composition is constant throughout the growth period to sexual maturity.

Statistical Analysis Regressions of mean AFE, WFE, and WAFE on generations were calculated in the selected lines from generation 0 to 54 for the AFE and from generation 27 to 54 for the last 2 traits. Divisions of generations into overlapping intervals corresponding to the propagation of the relaxed lines (6 to 14, 13 to 20, 19 to 27, 26 to 35, 34 to 44, and 43 to 54) provided regressions where selected and relaxed lines could be compared. Six groups of relaxed selection lines (initiated every 7 to 11 generations) were analyzed separately for each line (HWS, LWS) to compare linear slopes of selected vs. relaxed lines for mean AFE and 3 groups for WFE and WAFE. The results provided information

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on whether relaxation (cessation) of artificial selection at different stages of the selection experiment resulted in changes in regressions of the correlated response to selection for BW8. Prior to analysis, WFE was transformed to common logarithms. All calculations were conducted using regression procedures in JMP software (JMP 11, 2015). Default P-value = $P < 0.05$.

RESULTS AND DISCUSSION

Age at First Egg

Sexual maturity of females is defined here as the age when the first egg is oviposited (laid). There is a time lapse of approximately 26 h from ovulation to oviposition. A complete overview of the correlated response of AFE in the selected and their respective relaxed lines is seen in Figure 1. There was little change from the increase in HWS between the earlier and later generations. Moreover, the relaxed lines responded similarly to HWS from environmental factors throughout the experiment. The AFE increased in LWS throughout the experiment, while its relaxed lines generally reached sexual maturity at younger ages. Means and standard errors for age at sexual maturity were 212.64 ± 3.15 d in LWS and 180.57 ± 1.98 d in HWS during generations 0 to 54.

It is evident from Figure 1 that there were similarities and differences between the selected and their respective relaxed lines. To ascertain what occurred overall

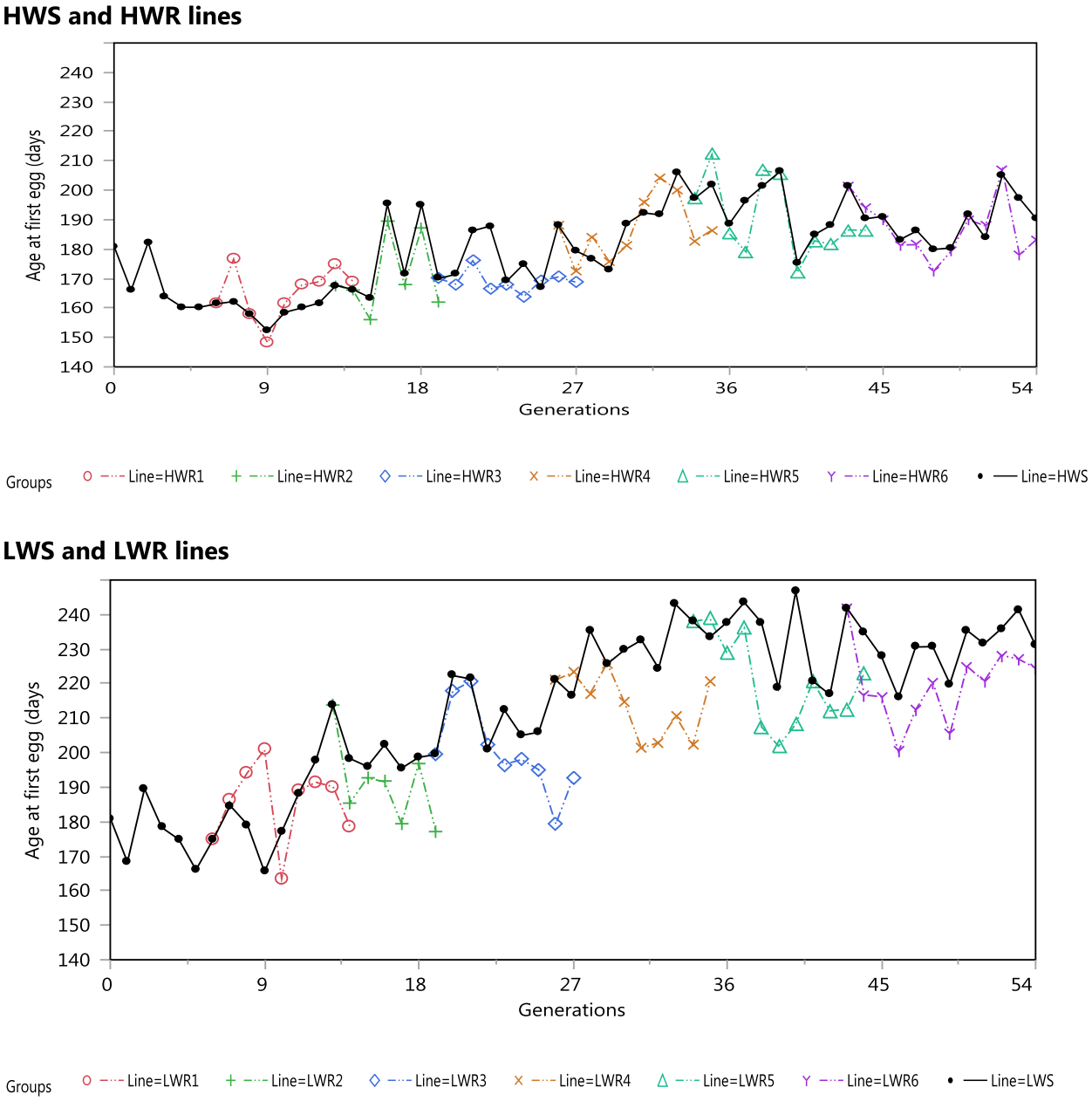


Figure 1. Mean age (d) at first egg for all lines across generations. HWS = high weight selected line; LWS = low weight selected line; HWR = high weight relaxed line; LWR = low weight relaxed line (numbers following letters for HWR and LWR relate to chronology of when selection was relaxed and correspond to S6, S13, S19, S26, S34, and S43).

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Table 1. Regressions of age (d) at first egg (AFE) on generation.

	Generation interval						
Line	0 to 54	6 to 14	13 to 20	19 to 27	26 to 35	34 to 44	43 to 54
HWS	0.63 ± 0.09*	0.86 ± 0.54	1.38 ± 2.04	0.61 ± 1.15	2.77 ± 0.80*	−0.79 ± 0.89	0.20 ± 0.70
HWR		1.11 ± 1.15	1.34 ± 2.53	−0.23 ± 0.45	1.43 ± 1.08	−1.78 ± 1.16	−0.55 ± 0.86
LWS	1.25 ± 0.10*	4.01 ± 1.35*	0.80 ± 1.57	0.62 ± 1.23	1.81 ± 0.70*	−0.75 ± 1.01	0.28 ± 0.66
LWR		0.15 ± 1.56	−3.56 ± 1.99	−3.32 ± 1.21*	−1.62 ± 0.91	−2.36 ± 1.07	0.39 ± 0.95

*Indicates regressions significantly different from zero ($P < 0.05$).
HWS = high weight selected line; LWS = low weight selected line; HWR = high weight relaxed line; and LWR = low weight relaxed line.

and during time periods, the regressions are presented in Table 1. The overall regression of AFE on generation (Gen 0 to 54) was 0.63 ± 0.09 and 1.25 ± 0.10 d for the HWS and LWS, respectively. Although HWS became sexually mature at a younger age than LWS, over the course of the selection experiment, there was a delay of 34 and 68 d in HWS and LWS, respectively. Thus, while both selected lines experienced delays in reaching sexual maturity, the delay was greater for LWS than HWS, suggesting a lower efficiency in body weight conversion concomitant to a decrease in feed intake with the direction of selection. When viewed within periods of selection, the regression of AFE on generation was positive and higher (4.01 ± 1.35 d) as early as the generation interval 6 to 14 for LWS, whereas response for HWS was also positive but lower (0.86 ± 0.54 d), indicating an earlier onset of sexual maturity.

During generation periods 13 to 20, 19 to 27, and 26 to 35, regressions were positive and higher in HWS than HWR. Although the positive response in LWS was similar to HWS, all regressions for LWR were negative, as they were maturing at younger ages than LWS. The differences between LWS and its relaxed counterparts became obvious as early as the 13 to 20 generation interval. The lack of difference between the selected and relaxed HW lines may be attributed to controlling the onset of lay by restricting feed intake (Liu et al., 1995).

During the generation interval 34 to 44, AFE decreased in both the selected and their respective relaxed lines, as shown by the negative regressions of AFE on generation (Table 1). Although none was significant, regressions were lower for the selected than their respective relaxed lines. This pattern suggests that there were environmental factors that caused the selected and relaxed lines to respond similarly during this period. It is also obvious that the effects of relaxing selection on AFE was greater in the low than high lines. This pattern was consistent with those reported by Dunnington and Siegel (1985) that relaxation of selection decreased AFE. The direction of response changed to positive in generations 43 to 54 for HWS, LWS, and LWR, but not HWR. The HW lines reached sexual maturity at younger ages than the LW lines, which was consistent with results reported by Liu et al. (1995) that the regression of AFE on generation was higher in LWS than HWS.

Body Weight at First Egg

For a complete overview of the correlated responses of WFE in the selected and their respective relaxed lines from generation 27 to 54, see Figure 2. WFE, which was not measured prior to generation 27, was greater in the HW than LW lines. HWS commenced egg production at a heavier BW than its relaxed lines and LWS commenced lay at a lower BW than its relaxed lines. WFE ranged from 2,735 to 3,291 g in HWS and 1,078 to 1,329 g in LWS. The overall means and standard errors for WFE were 2,939 ± 26 g for the HWS and 1,183 ± 14 g in the LWS, a 2.5 fold difference in WFE between them. Similarities in both lines included declining WFE from generation 27 to 44, which increased thereafter. The increase, however, was greater in the HWS than LWS concomitant to direction of selection.

Regressions of WFE on generation 27 to 54 were 9.82 ± 2.70 g and −5.20 ± 1.27 g for HWS and LWS, respectively (Table 2). The increase in WFE in the HWS was greater during the last generation interval (generation 43 to 54) than the 2 previous intervals where the regressions of WFE on generation were negative. The difference may be attributed to the feed restriction and its effects on body weight after selection age. In contrast, the decrease in WFE for LWS was influenced more during the 2 earlier intervals (generations 27 to 35 and 34 to 44) where the regressions of WFE on generations were negative other than positive during the last generation interval. The magnitude of decrease was higher in LWS than HWS during the 2 earlier intervals. In contrast, regression coefficients for the last generation interval were higher in HWS than LWS (30.02 ± 8.94 vs. 0.48 ± 4.57 g) indicating an increase in WFE. Numerous factors may come into play in correlated responses during long-term selection experiments. Here we examined generational intervals. As described by Jambui et al. (2017), over time there were mutations (Alexander at al., 2015), proviral inserts (Ka et al., 2009), and maternal and nonadditive genetic effects (Pettersson et al., 2011). Additionally, some standing residual genetic variation from the base population (Sheng et al., 2015) could come into play with adjustments to feed intake and subtle changes in the environment.

For selected and relaxed lines, regressions of WFE on generation during the first 2 generation intervals (27 to 35, 34 to 44) were decreasing in both the HWS and

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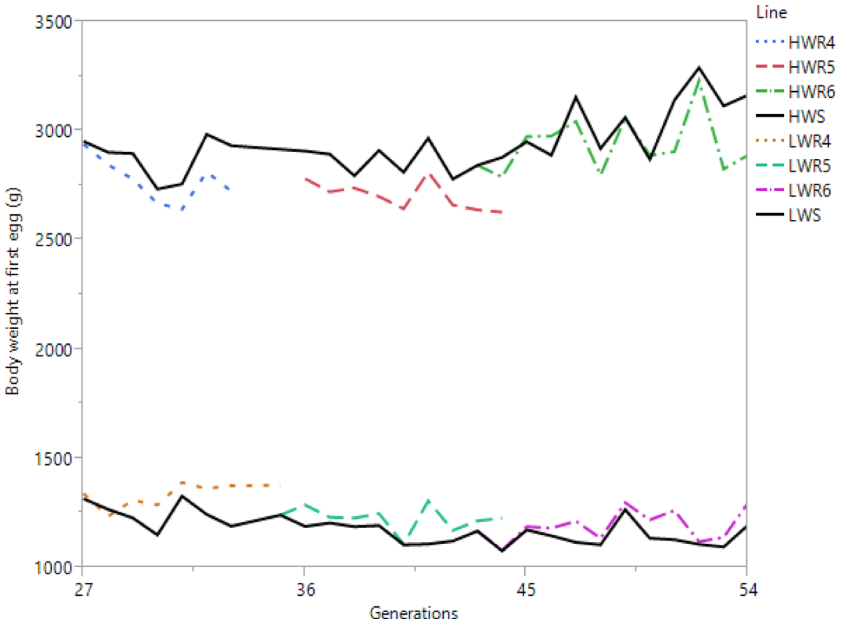


Figure 2. Mean body weight (g) at first egg for all lines across generations. HWS = high weight selected line; LWS = low weight selected line; HWR = high weight relaxed line; LWR = low weight relaxed line (numbers following letters for HWR and LWR relate to chronology of when selection was relaxed and correspond to S26, S34, and S43).

Table 2. Regressions of body weight at first egg (g) on generation.

	Generation interval			
Line	27 to 54	27 to 35	34 to 44	43 to 54
HWS	9.82 ± 2.70*	−1.21 ± 20.06	−4.05 ± 8.44	30.02 ± 8.94*
HWR		−30.36 ± 16.80	−15.06 ± 7.07	7.24 ± 11.10
LWS	−5.20 ± 1.27*	−6.95 ± 8.64	−14.35 ± 3.58	0.48 ± 4.57
LWR		13.14 ± 6.22	−5.15 ± 6.28	7.23 ± 5.66

*Indicates regressions significantly different from zero ($P < 0.05$).
HWS = high weight selected line; LWS = low weight selected line; HWR = high weight relaxed line; and LWR = low weight relaxed line.

HWR, although WFE was lower in the latter than former. Although HWR commenced egg production at a lower BW than HWS, both responses increased during the last generation interval (generation 43 to 54) with the response significant for HWS but not HWR (30.02 ± 8.94 vs. 7.24 ± 11.10). The similarity in response on the intervals for HWS and HWR demonstrates that feed restriction practiced in the high weight lines influenced the pattern of response.

In LWS, there was a similar trend of decreasing WFE in the earlier generations before increasing during the last generation interval. This increase in WFE during generation 43 to 54 occurred in both LWS and LWR. Selection for low BW has resulted in a threshold BW below which pullets do not reach sexual maturity. For the LWS, the minimum of 1,000 g, below which pullets did not commence egg production (Dunnington et al., 1983), continues to the present. This threshold may be due to anorexia in this line because by force-feeding, non-layers reached the threshold body weight necessary for the attainment of sexual maturity as measured by producing eggs (Zelenka et al., 1988). Jungle fowl, bantams, and dwarf chickens commence lay at less than this threshold weight.

When selection was relaxed, WFE was lower for the HWR than HWS. For the low-weight lines, LWR was heavier than LWS at sexual maturity (Figure 2). Thus, there was an opposition between the forces of artificial and natural selection where selection increased WFE in HWS and decreased WFE in LWS, implying that minimum WFE was line (population) specific.

Ratio of WFE and AFE

The correlated response of WAFE in the selected and their respective relaxed lines from generation 27 to 54 is seen in Figure 3. WAFE was higher in HW than LW lines, reflecting greater growth to sexual maturity in HWS with differences of >3-fold in the selected lines. WAFE in HWS decreased during the earlier generations and increased during later generations. In contrast, WAFE in LWS decreased throughout. Upon comparing the relaxed and selected lines, WAFE was generally greater in the selected than in relaxed HW lines. In contrast, WAFE was consistently higher in LWR than LWS. Upon relaxation of selection, there was a decrease in WAFE for HWR and an increase in LWR. Similarities and differences between the selected and their respective relaxed lines for WAFE are evident in Figure 3, and regressions of WAFE that occurred overall and during time periods are presented in Table 3. The regressions of WAFE from generation 27 to 54 were 0.03 ± 0.02 g/d and −0.02 ± 0.01 g/d in HWS and LWS, respectively. Means and standard errors for daily weight gain from hatch to sexual maturity were 15.60 ± 0.19 and 5.15 ± 0.08 g/d in HWS and LWS, respectively.

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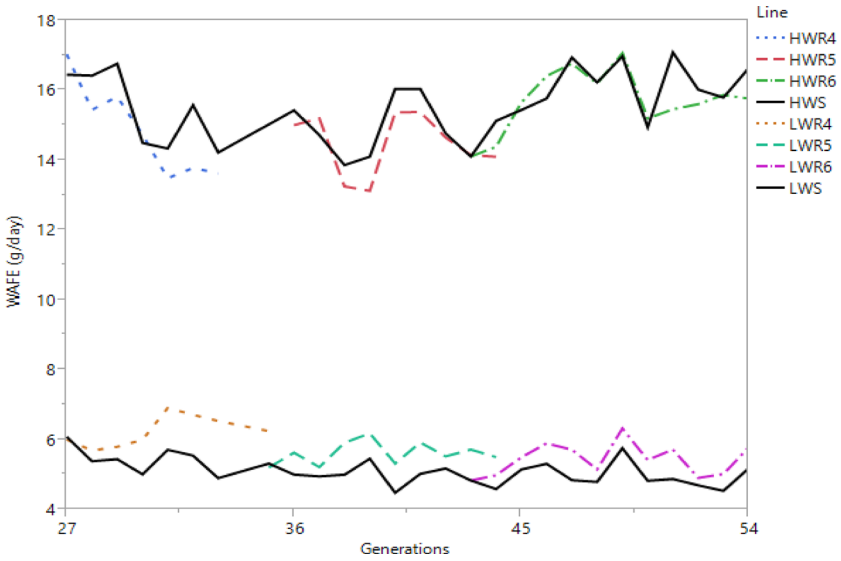


Figure 3. Mean ratio of body weight and age at first egg (WAFE) for all lines across generations. HWS = high weight selected line; LWS = low weight selected line; HWR = high weight relaxed line; LWR = low weight relaxed line (numbers following letters for HWR and LWR relate to chronology of when selection was relaxed and correspond to S26, S34, and S43).

Table 3. Regressions of WAFE (g/d) on generation.

	Generation interval			
Line	27 to 54	27 to 35	34 to 44	43 to 54
HWS	0.03 ± 0.02	−0.39 ± 0.15	0.01 ± 0.11	0.14 ± 0.07
HWR		−0.57 ± 0.11*	−0.03 ± 0.12	0.08 ± 0.07
LWS	−0.02 ± 0.01*	−0.07 ± 0.05	−0.05 ± 0.03	−0.004 ± 0.03
LWR		0.10 ± 0.06	0.02 ± 0.04	0.02 ± 0.04

*Indicates regressions significantly different from zero ($P < 0.05$).
HWS = high weight selected line; LWS = low weight selected line; HWR = high weight relaxed line; and LWR = low weight relaxed line.

Although the difference in WAFE between the selected lines was >3 fold, there was a small but consistent difference, with HWR being generally lower than HWS. This may be due to the restriction of their feed intake after 8 wk of age and thus a steady growth to reduce reproductive complications. The difference in patterns between HW and LW lines shows that the HW lines were more variable in response than the LW lines, which were decreasing overall. In contrast, WAFE was consistently higher in LWR than LWS, thus delaying the onset of sexual maturity in the latter. The difference in WAFE between relaxed and selected lines was greater and more obvious from the beginning in the LW than the HW lines (Figure 3).

That the regressions of WAFE on generations were negative throughout all generation intervals in LWS (Table 3) suggested slower growth to maturity than LWR, which was always positive. The importance of WAFE (assuming consistent body composition) is that it can be used to evaluate whether or not pullets are gaining sufficient BW to reach sexual maturity. This reasoning is consistent with commercial breeder guides, which recommend feeding broiler breeders to a target weight during the growing period. A WAFE lower than the threshold ratio, such as ratio of <4 g/d in LWS, may be an indication that pullets in these lines are unable to commence lay and may be anorexic. Feeding programs such as force-feeding have been successful in bringing non-laying pullets with lower WAFE into egg production. Namely, anorexia was restricting them from reaching threshold BW required for sexual maturity (Zelenka et al., 1988).

From a selection point of view, this growth information may be monitored at young ages, because most likely an individual with a lower body weight gain as early as 4 to 8 wk of age may not consume sufficient feed to meet the threshold BW to commence lay. Namely, WFE is more closely tied to adult than to juvenile BW (Dunnington et al., 1983). A low but positive genetic correlation between WFE and early growth suggests that continued selection for rapid growth should produce only minor further increases in WFE (Soller et al., 1984). Thus, feed restriction practiced in HW chickens during the rearing period will further control growth and keep BW at a constant requirement for sexual maturity and reproductive success. A WAFE higher than the target may indicate that the chicken is obese with possible reproductive complications such as erratic ovulation and defective egg syndrome (**EODES**) (Jaap and Muir, 1968; van Middlekoop, 1971, 1972; van Middlekoop and Siegel, 1976), as well as associated metabolic and cardiovascular disorders. Selection for increased BW has resulted in heavier WFE and negative effects on reproductive performance in adults when fed ad libitum. Early maturity associated with increased juvenile growth has been associated with reproductive problems, such abnormal hierarchies in egg formation (Hocking, 2004). Accordingly, from a practical aspect, human intervention is realistic in monitoring the growth of breeders.

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The relationship between age and BW showed that there are minimum thresholds for ages and BW at which the chickens cannot reach sexual maturity, as a correlated response to selection for BW8 (Dunnington et al., 1983). The WAFE showed there is a minimum average daily growth at which sexual maturity will occur. Below that, LWS pullets will not commence egg production. For the HWS pullets, a WAFE above the minimum of approximately 15 to 17 g/d contributes to obesity and associated reproductive complications. Anorexia associated with selection for low BW reduces growth and delays or precludes sexual maturity. Thus, in the commercial world of feeding programs for breeders, be they for egg or meat production stocks, we are dealing with population specifics to achieve the target weights optimum for high reproductive rates.

General

Long-term divergent selection for BW8 resulted in changes in AFE, WFE, and WAFE. Although sexual maturity was delayed in both selected lines, it was more pronounced in LWS than HWS. Selection for low BW resulted in correlated decreases in WFE and WAFE. Sexual maturity declined more in the LWS than the HWS lines. Selection for heavier body weights resulted in higher WFE and WAFE, and the feed restriction program may have influenced responses.

Minimum AFE, WFE, and WAFE in relation to sexual maturity were specific for each line and influenced by selection for BW8. An important measure for assessing growth to reach sexual maturity was WAFE. This measure provides a "yardstick" to which sexual maturity is likely or unlikely to occur and the possibility of reproductive complications or a case of anorexia. There are likely economic advantages for this measure from the point of view of selection experiments or commercial breeding programs. Lastly, an opposition between relaxed and artificial selection is seen in the difference in response between selected and relaxed lines. Namely, there is enhanced sexual maturity and greater fitness with relaxed than artificial selection on the high and low extremes of body weight.

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